Machine learning for sustainable process modelling

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ML for real-time control of copper heap leaching

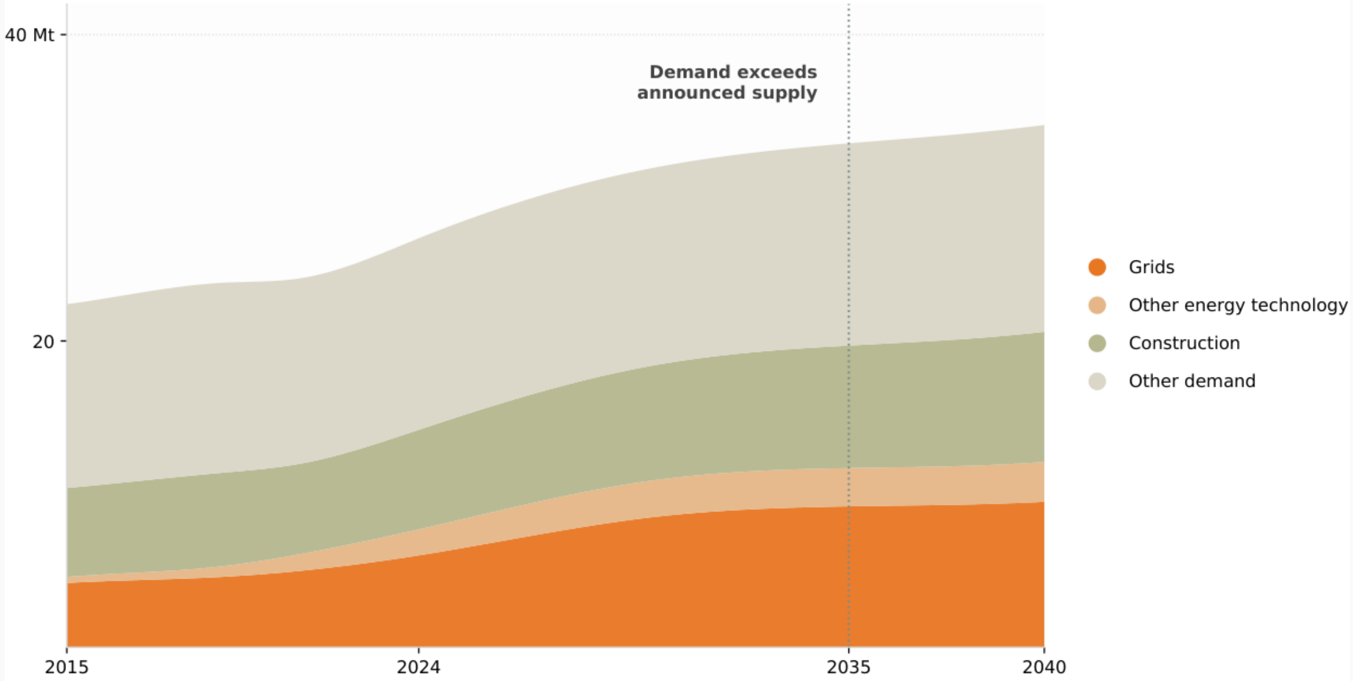
Goal:
ML model that makes copper heap leaching controllable in real time, improving recovery rates from low-grade ores.

Problem statement:
Available heap leaching controllers are accurate but too slow for real-time control at full heap scale.

Proposed solution:
Replace controller's slow internal predictor with a neural surrogate for an optimised irrigation in real time.

Copper supply gap

Demand outrunning supply and ores are becoming leaner



Global copper demand to 2040 by sector, with supply outlook[1]

-40%
Ore grades since **1991**[2]

30%
Supply gap by **2035**[1]

27 ⟶ 37 Mt
Demand increase by **2050**[1]

Copper demand is forecasted **to outrun** supply by **2035**. With few new deposits, copper must be sourced from low-grade ores.

Heap leaching is the most established route for low-grade ores and making its efficiency a sustainability priority **ensures meeting present demand without compromising future supply for upcoming generations**.

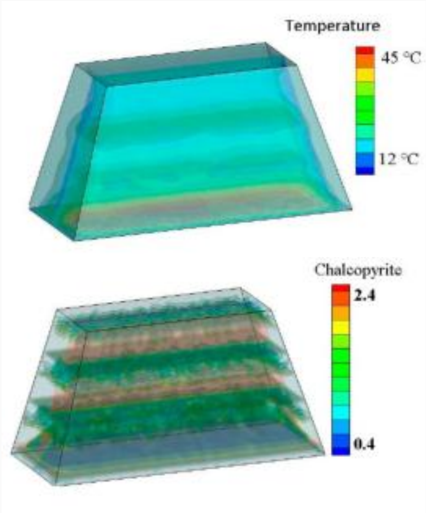
UCL

[1] International Energy Agency, Global Critical Minerals Outlook 2025.
[2] Calvo et al., Decreasing ore grades in global metallic mining, Resources, 2016.

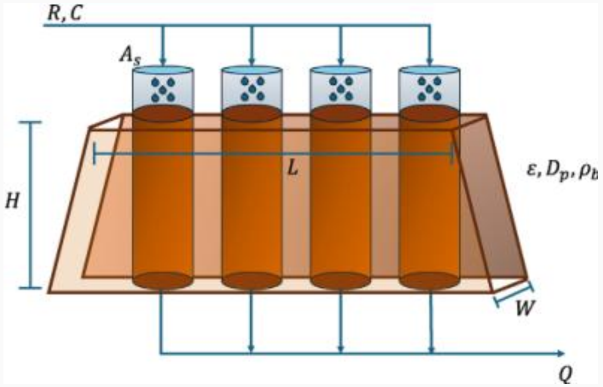
Heap is a black box

Large, heterogeneous, and unmeasurable during operations

Copper heap leach site in Chile[3]



CFD heap cross-section showing internal heterogeneity[4]



Heap discretised into 1D columns[7]

Heaps **span kilometres**[3] and **stands 5-15 m** high making internal state inaccessible to direct measurement during operation horizon.

Uniform surface irrigation creates **preferential flows**[4] and **dry zones** within the heap resulting in uneven and suboptimal recovery.

To model the heap, it is **split into vertical columns**, each solved as a 1D problem.

[3] BHP, 2024 Chilean copper site tour [Presentation], Minerals Americas, 2024.
[4] McBride et al., Heap leaching: Modelling and forecasting using CFD technology, Minerals, 2018.
[7] Olivares et al., Saturation regulation in heap leaching: a nonlinear MPC approach, Minerals Engineering, 2025.

Surveying the field

Approaches assessed against the heap leaching control problem

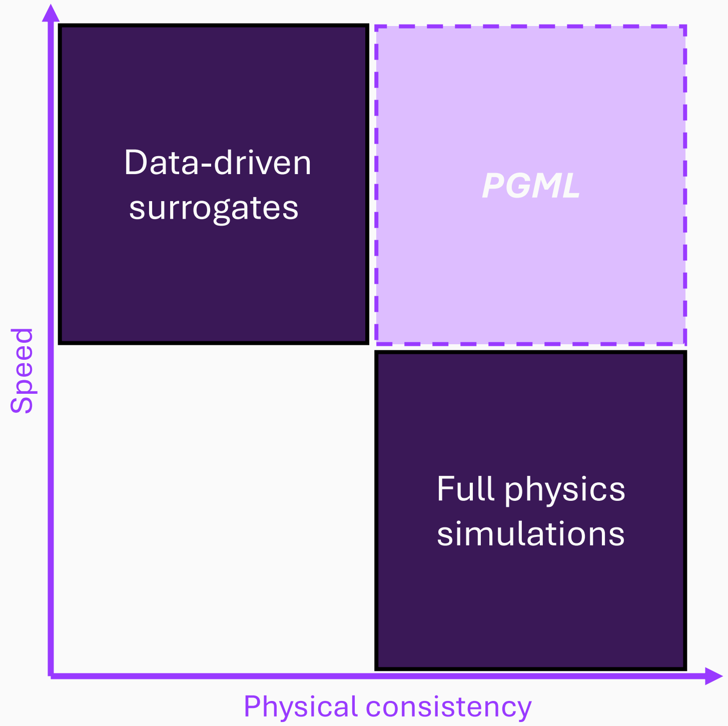
Reference	Modelling approach	Control strategy	Main limitation	Relevant to heap leaching control
Liu & Hashemzadeh[5]	Empirical	Open loop	No dynamic feedback	Low
McBride et al.[4]	CFD	None	Computationally prohibitive	Very low
Canales et al.[6]	Deep learning (DRL)	AI agent	No physical constraints	Medium
Olivares et al.[7]	First principles	NMPC (feedback)	Slow at full scale	Medium
Eppink & Briesen[8]	Single shooting	Trajectory planning	No dynamic feedback	Medium
This thesis	**Physics-guided ML**	**PGML-MPC**	**Training effort required**	High

[4] McBride et al., Heap leaching: Modelling and forecasting using CFD technology, Minerals, 2018.
[5] Liu & Hashemzadeh, Solution flow behavior in heap leaching, Hydrometallurgy, 2017.
[6] Canales et al., Control of heap leach piles using deep reinforcement learning, Minerals Engineering, 2024.
[7] Olivares et al., Saturation regulation in heap leaching: a nonlinear MPC approach, Minerals Engineering, 2025.
[8] Eppink & Briesen, Optimal control of infiltration conditions in heap leaching, Minerals Engineering, 2026.

A gap in existing methods

Surveying approaches in literature to assess the gap

- **Full physics simulations** capture the physics but cannot run in real time due to being **computationally expensive**. [4]
- **Data driven surrogates** are fast but **physically unconstrained** and extrapolate poorly. [7]
- Available methods are either **fast or physically consistent** but rarely both at heap scale.



[4] McBride et al., Heap leaching: Modelling and forecasting using CFD technology, Minerals, 2018.
[7] Olivares et al., Saturation regulation in heap leaching: a nonlinear MPC approach, Minerals Engineering, 2025.

Project objectives

What the surrogate must achieve

Speed

Sub-second prediction per column allowing for real-time control

Physical consistency

Predictions remain physically sound, not merely fit to data

Scalability

Fast enough per column predictions to model a full 3D heap in real time

Generalisation

Accuracy held across varied soil conditions, not tailored for one fixed case

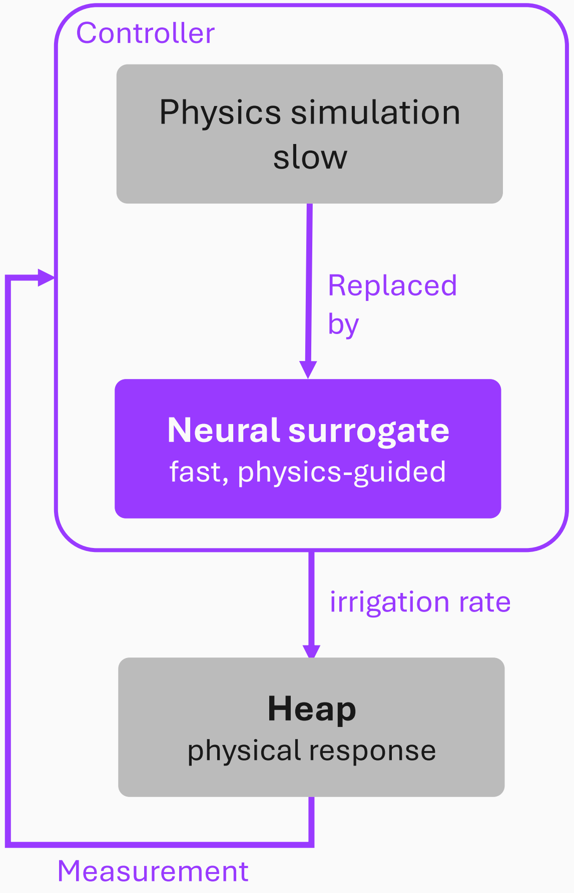
Proposed solution

Guided by the controller's own physics, not generic data-fitting

Bottleneck :
The controller predicts the heap's response before choosing an irrigation rate. This prediction is a physics simulation, and it is the bottleneck.[7]

Swap:
Replace that simulation with a neural surrogate that predicts the same response in a fraction of the time.

Novelty:
The surrogate is trained to obey the controller's own physical model, so it remains physically consistent rather than a black box.



[7] Olivares et al., Saturation regulation in heap leaching: a nonlinear MPC approach, Minerals Engineering, 2025.

Proposed solution

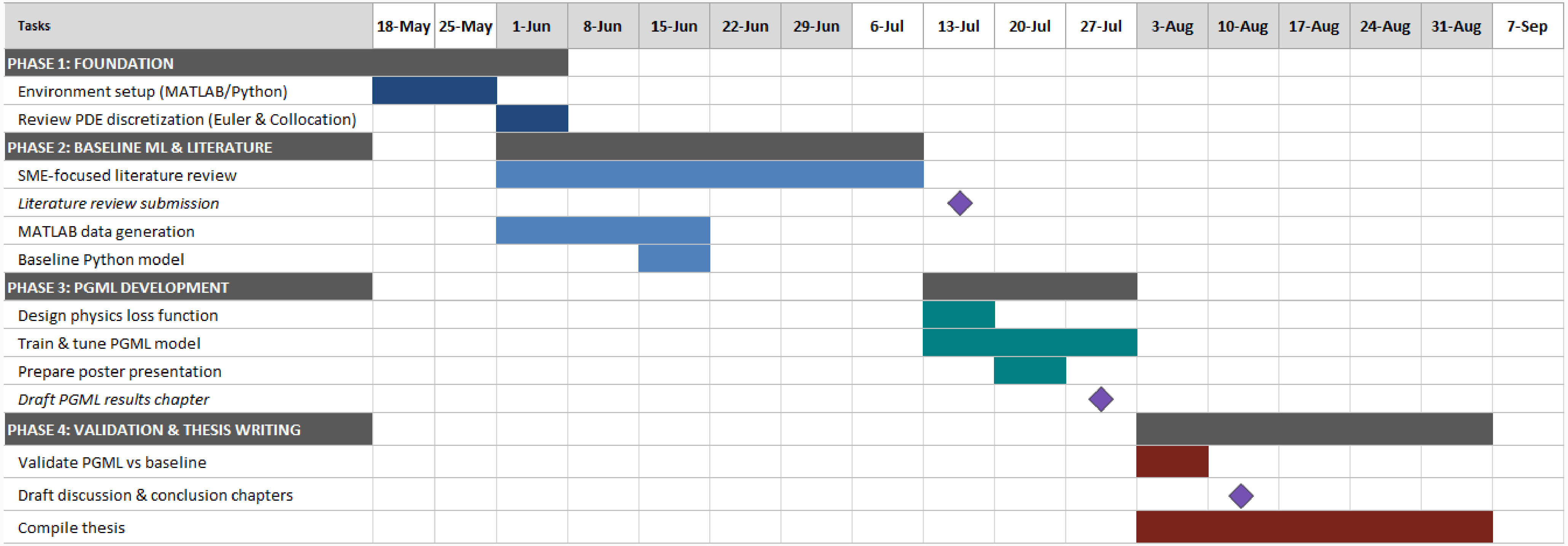
Contribution mechanism

Conventional controller (Olivares et al., 2025)	**Physics-guided ML (PGML)** (Proposed solution)
Forward model predict heap one step ahead	**Forward model** one surrogate pass
Identify / re-fit retuning of soil parameters 1.4-3.5 s	~~**Identify / re-fit**~~ PGML eliminates the slowest step
Optimise search for best irrigation rate 0.06-0.10 s	**Optimise** grid search
Apply + loop actuate and feedback loop	**Apply + loop** actuate and feedback loop
Duration: < 5 seconds Per column	**Duration: < 1 second** Per column

[7] Olivares et al., Saturation regulation in heap leaching: a nonlinear MPC approach, Minerals Engineering, 2025.

Path forward and conclusions

Timeline, milestones, and remaining work to submission



Progress:
- Gantt chart completed
- Literature surveyed
- Preliminary pipeline built

Next steps:
- Finalise literature review
- Generate and train on varied soil data
- Evaluate generalisation
- Thesis write-up